Polymeric Chemistry
DSC and TGA

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1 Theory Part

1.1 DSC

The DSC (Differential Scanning Calorimetry) uses the principle of the isotherm calorimetry. The Sample and a references get heated in two separate crucibles. The absorbed energy is measured, and in case of different absorption the one with less gets more power while this added power is being measured and plotted against time (t). The Integral of dW/dt is the difference between the sample and the reference, also shown in picture 1 is the Integral underneath the curve which defines the smelting enthalpy which is calculated by using equation (1). This curve is influenced by the spectrum of lengths the polymer provides, meaning if the polymer has many chains with different lengths the peak gets wider. The reference is always a known Sample or something, for which the heat capacity is known. In our case the reference was the empty crucible.

$$\Delta H_{\mathrm{m}} = \Delta H_{\mathrm{Sample}} - \Delta H_{\mathrm{Reference}} \tag{1}$$

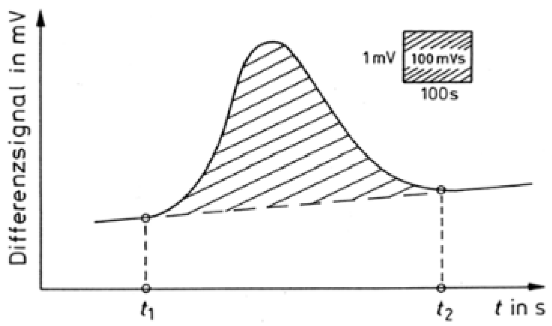

Figure 1: This picture shows the thermogram of a Polymer, the differential signal in [mV] is plotted against time in [s]. [1]

Via this melting enthalpy the crystallization grade can be determined by using the melting enthalpy and by using ΔH^0_{m}. ΔH^0_{m} cannot directly be determined, because polymers are not fully crystalline. Therefore, a extrapolation is used to determine the ideal case. Meaning the length of the polymer is imagined to be infinite. All of this results in following equation.

$$\alpha = \frac{\Delta H_{\mathrm{m}}}{\Delta H^0_{\mathrm{m}}} \tag{2}$$

1.1.1 Glass transition temperature (T_g) and melting temperature (T_m)

The Glass transition temperature (T_g) is the temperature range where a complete coupled segment movement of chains (10-15 chain links) is possible. These chains were frozen before that temperature and can only be defined if the polymer is amorphous. This can also be seen in a change of the heat capacity.

The melting temperature (T_m) is only present in semi crystalline polymers. These have a lamellar structure (folding crystal) with amorphous parts between the backfolding parts. When reaching the melting temperature these backfolded parts are moving apart. The melting temperature can be influenced by the amount of amorphous parts in the polymer, meaning if there are more amorphous parts the melting temperature gets higher. It is also influenced by the thickness of the crystall, this is also shown by the following equation:

$$T_{\mathrm{m}} = T_{\mathrm{m}}^{0} \cdot \left(1 \cdot \frac{2\sigma}{\rho_c \cdot \Delta H_{\mathrm{m}}^{0} \cdot l}\right) \tag{3}$$

In this equation σ is the interfacial tension of the fin surface area, $\Delta H_{\mathrm{m}}^{0}$ is the specific melting enthalpy, l is the crystall thickness and ρ_c is the density of the crystalline area. T_{m}^{0} is the melting temperature for an ideal crystall and is defined by:

$$T_m^0 = \frac{\Delta H_m^0}{\Delta S_m^0} \tag{4}$$

Both of these temperatures can be analyzed and read out of the thermogram. For semi crystalline polymers with a bigger amorphous share the peak is moved to the right because of the higher temperature.

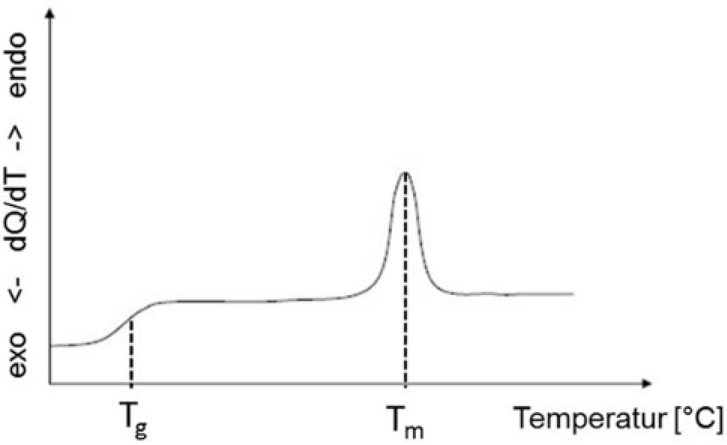

Figure 2: This picture shows an example of a thermogram of a polymer, in which T_g is the Glass transition temperature and T_m is the melting temperature.

1.2 TGA

The main principle of TGA (Thermo gravimetry) is the heating of a Sample to the point where it gets destroyed. In the process the mass is monitored and plotted against temperature. It is also possible to plot the temperature against the time. If these two graphs are compared a clear corelation can be observed.
For analytical purposes the m-T-curve ist the more valuable one, because from this curve things like the starting temperature and the end temperature of the degrading can be determined by tangent constructions. This can also be observed in picture (3).

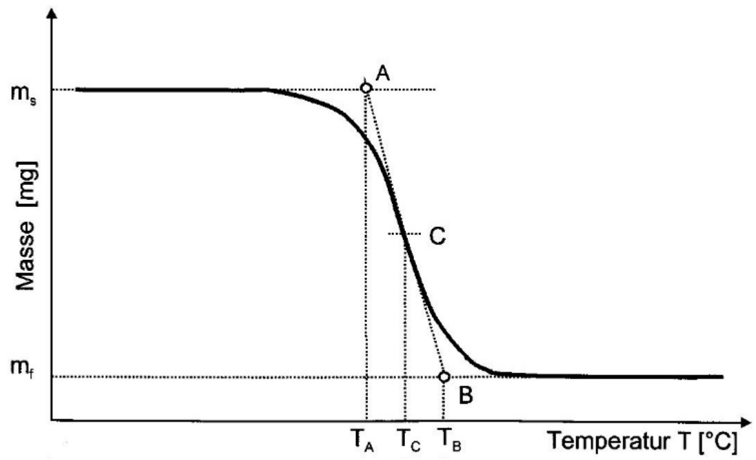

Figure 3: The figure shows the m-T-curve for a polymer. Also shown in the picture are the points A (starting point), B (end point) and point C (middle point) and the three correlating temperatures.

Via this diagram the material specific properties can be characterized.

2 Results

Two different samples were prepared, and analyzed, using DSC and TGA. The Data of the samples (PET and *it*-PS) are listed in the Tables 1 and 2 below.

Table 1: Listed are the meltingtemperatures (T_{m}), glass temperatures (T_{g}) and the melting enthalpy (Δ $\mathrm{H_m}$) from the *it*-PS and PET heating curves.

Sample	Δ $\mathrm{H_m}$ [$\frac{J}{g}$]	$\mathrm{T_m}$ [°C]	$\mathrm{T_g}$ [°C]
PET 1.hc	38.368	249.85	-
PET 2.hc	32.360	246.00	84.36
it-PS 1.hc	24.194	221.61	99.69

Table 2: Listed are the crystallizationtemperature (T_{c}) and the crystallization enthalpy (Δ $\mathrm{H_c}$) of the coolingcurve of PET.

Sample	Δ $\mathrm{H_c}$ [$\frac{J}{g}$]	$\mathrm{T_c}$ [°C]
PET cc	-34.563	182.73

As shown in the appendix on pages 9 and 10, for *it*-PS were no crystallization enthalpy and no second melting enthalpy observed. Thus, before the measurement both polymers are semi-crystalline, however, the chosen cooling conditions of 20 $\frac{°\mathrm{C}}{\mathrm{min}}$ are too fast for the PS to crystallize again and is therefore amorphous during the measurement of the second heating curve. In literature the crystallization of *it*-PS is described as challenging and still studied. They also described the influence of different solvents and cooling rates. As an Exemple they often use cooling conditions of 10 $\frac{°\mathrm{C}}{\mathrm{min}}$. [2]
A signal can also be observed in the second heating curve of the *it*-PS, which occurs immediately

after the glass transition temperature. This may be due to the fact that, during rapid cooling, the chains are aligned parallel to one another, as shown on the left in Figure 4, but are not crystalline.

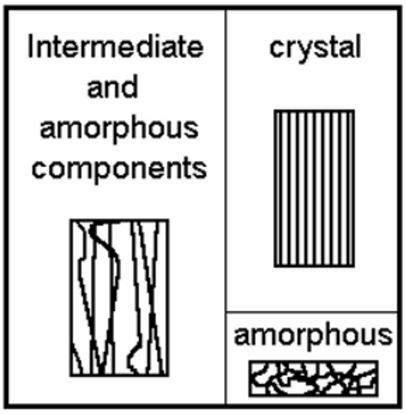

Figure 4: The figure shows possible structural elements of the polymer chains. [3]

Upon reaching the glass transition temperature, the chains move and can align in a partially crystalline manner. This alignment triggers a melting process, which, occurs at low temperatures. This is because the length of the crystalline regions is very small, and according to equation (3), the melting temperature is very low. [3]
Using the melting enthalpy, the degree of crystallization α were calculated to

$$\alpha_{\mathrm{PET}} = \frac{38{,}368\,\frac{\mathrm{J}}{\mathrm{g}}}{140\,\frac{\mathrm{J}}{\mathrm{g}}} = 0.274$$

and

$$\alpha_{it\text{-PS}} = \frac{24{,}194\,\frac{\mathrm{J}}{\mathrm{g}}}{87\,\frac{\mathrm{J}}{\mathrm{g}}} = 0.278.$$

For PET the curves can be seen on pages pages 11 to 13. The first curve shows the first heating curve, with an intens and sharp peak at the melting temperature. This is due to the fact that the crystalline parts have roughly the same length. The second curve shows the first cooling curve, with again an sharp and intense crystallization peak. For the last curve, being the second heating curve, we can observe a smaller and broader peak than before. This means that the crystalline parts have a broader spectrum of lenghts. Another interesting observation is a little plateau in the melting enthalpy, which is due to an higher amount of same lengths with this melting temperature. An other interesting part is the fact that the glass transition temperature is only visible in the second heating curve, meaning the pure PET sample we measured wasn't in the glass state but somehow been in the elastic state. This can be due to electromagnetic radiation or thermal energy that started the movement of the amorphous phase. With the cooling process in the cuvette the polymer got in the glass state, which is the reason for the observation of the glass transition temperature in the second heating curve. [1]

The last spektra is the TGA curve of PET, which can be seen on page 14. From 30 to roughly 400 °C the mass stays constant and the polymer stays stable. Afterwards the mass rapidly decreases with increasing temperature, which is due to depolymerization. In the end the mass decreased by 86 % and the remaining 14 % were observed as black powder being pure carbon. Due to this measurement the PET can be used up to 451,64 °C without significant depolymerization.

3 Questions

3.1 Question 1

The number indicates how many monomers are needed to make one full 360° rotation. The index says how many 360° rotations are needed until the order gets repeated.
For example in the case of 8_5 Polybutene. It needs 8 monomers to complete one 360° rotation and 5 full rotation until the order starts from the start.

3.2 Question 2

it-PP can be made by using the Ziegler-Natta-catalysis. In this case catalysts like $TiCl_3$ on $MgCl_2$-carrier are used in combination with aluminumalkenes. It can also be done via Metallocene-catalysis, this technique is specially used to control the tacticity and results in high graded *it*-PP.
it-PS can also be synthesized using the Metallocene-catalysis. But can also be synthesized via Anionic-Polymerization.

3.3 Question 3

Commonly the degree of crystallization decides how stiff a polymer is. The higher the degree of crystallization, the harder and stiffer the polymer. It also gets less ductile and more dense because crystalline parts can be packed better than amorphous parts.

3.4 Question 4

The Polymer is divided in different sections a back folding section(red), which would be the PE in our case and a amorphous section(black), which in our case would be the PS sections.

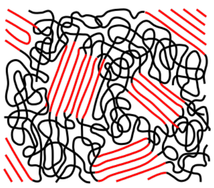

Figure 5: The picture shows the part-.crystalline

3.5 Question 5

To get an semi-crystalline polymer out of an polymerglass it has to be reheated. By reheating the polymer the molecular mobility gets high enough to form crystals. This process is also known as cold crystallization.

3.6 Question 6

The increase in mass can be explained by oxidation. While increasing the temperature in the TGA the polymer oxidizes, the added Oxygen increases the mass of the polymer (molecule if the polymer gets destroyed).

3.7 Question 7

The Mass degeneration curves for different heating procedures are shown in Figure 6.

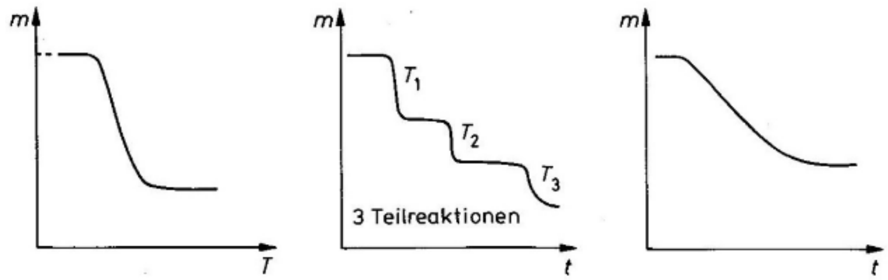

Figure 6: The diagramm on the left visuallizes the mass plotted against the temperature. The other two show the mass plotted against time. [1]

For an constant temperature gradient the graph on the left visuallizes the mass degeneration. For an isotherm measurement the right graph visuallizes the course and the middle one visuallizes an course for an stepwise isothermal procedure. [1]

3.8 Question 8

Due to the residual moisture, the PET decomposes to Ethylenglycol and Terephthalic acid. The Ethylenglycol eliminates water which results in vinylalcohol, which can tautomerize to Ethanal.

4 Appendix

Filename: C:\Users\DSC\Desktop\MD\G...\ps-iso-dsc.d6d
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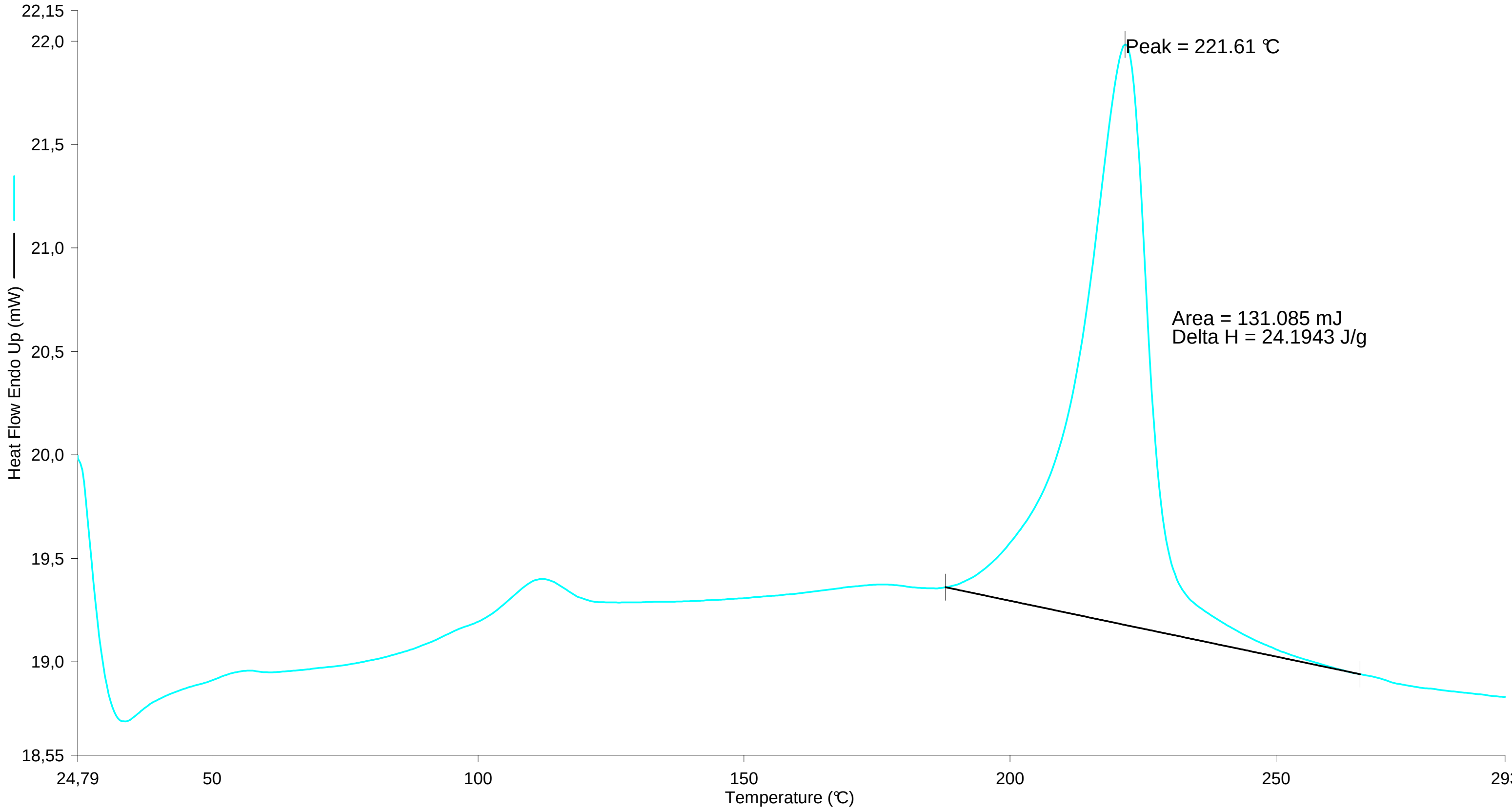
Comment:



10.03.2026 11:25:16

1) Heat from 30.00℃ to 300.00℃ at 20.00℃/min
2) Cool from 300.00℃ to 30.00℃ at 20.00℃/min
3) Heat from 30.00℃ to 300.00℃ at 20.00℃/min

Filename:	C:\Users\DSC\Desktop\MD\G...\ps-iso-dsc.d6d
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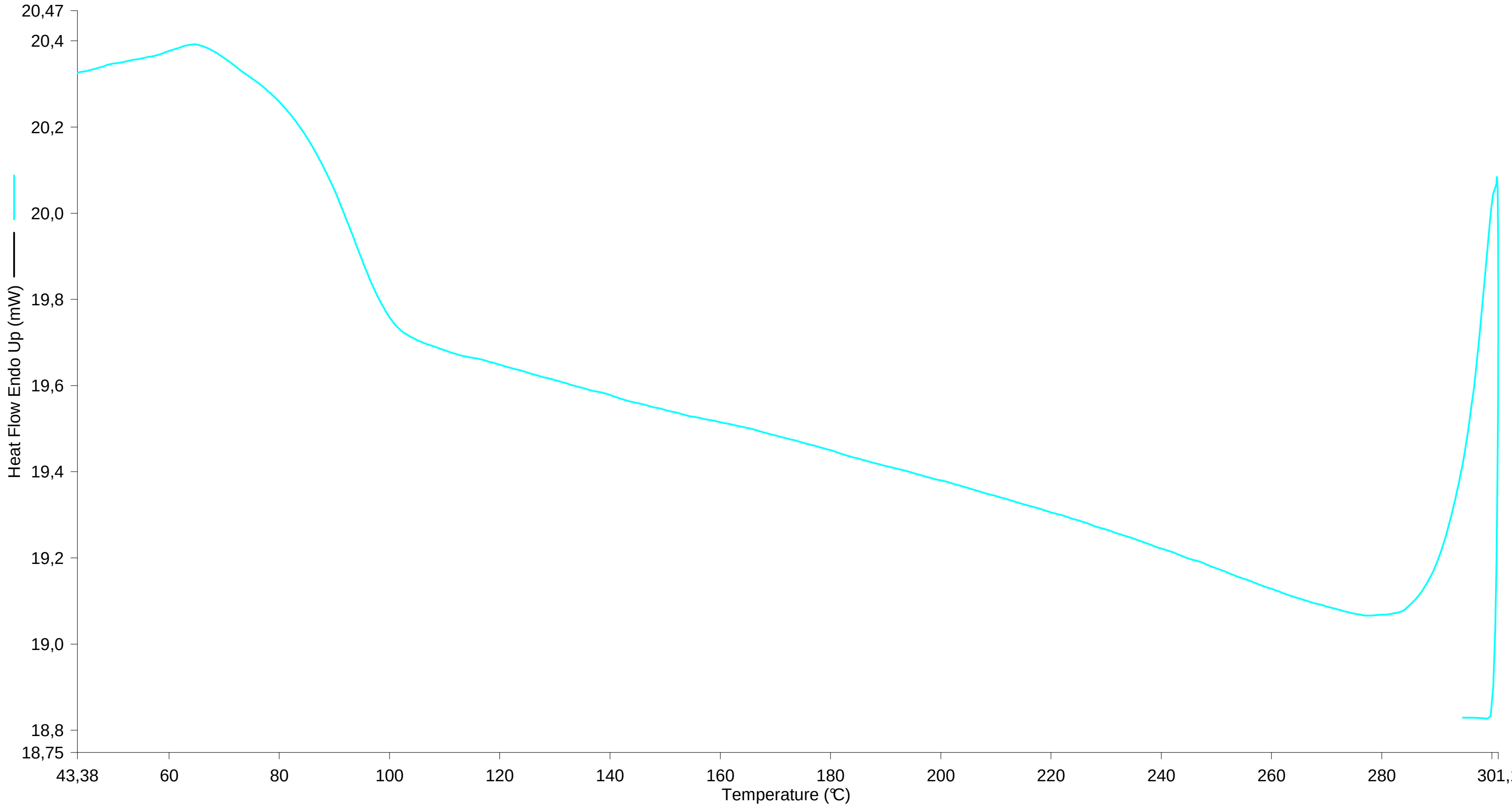
Sample Weight:	5.418 mg
Comment:	



10.03.2026 11:26:12

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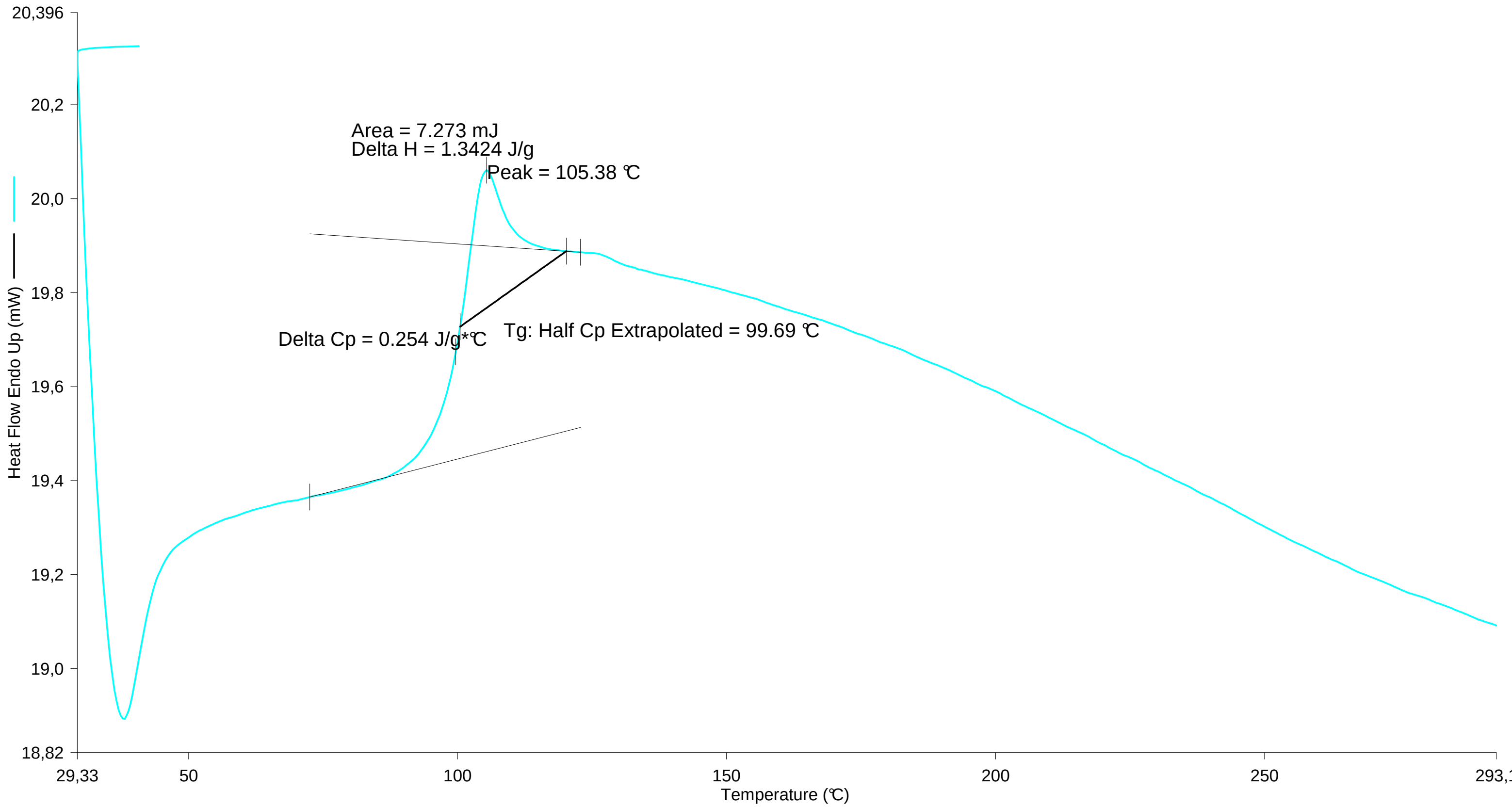
Sample Weight:	5.418 mg
Comment:	



10.03.2026 11:27:53

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3) Heat from 30.00℃ to 300.00℃ at 20.00℃/min

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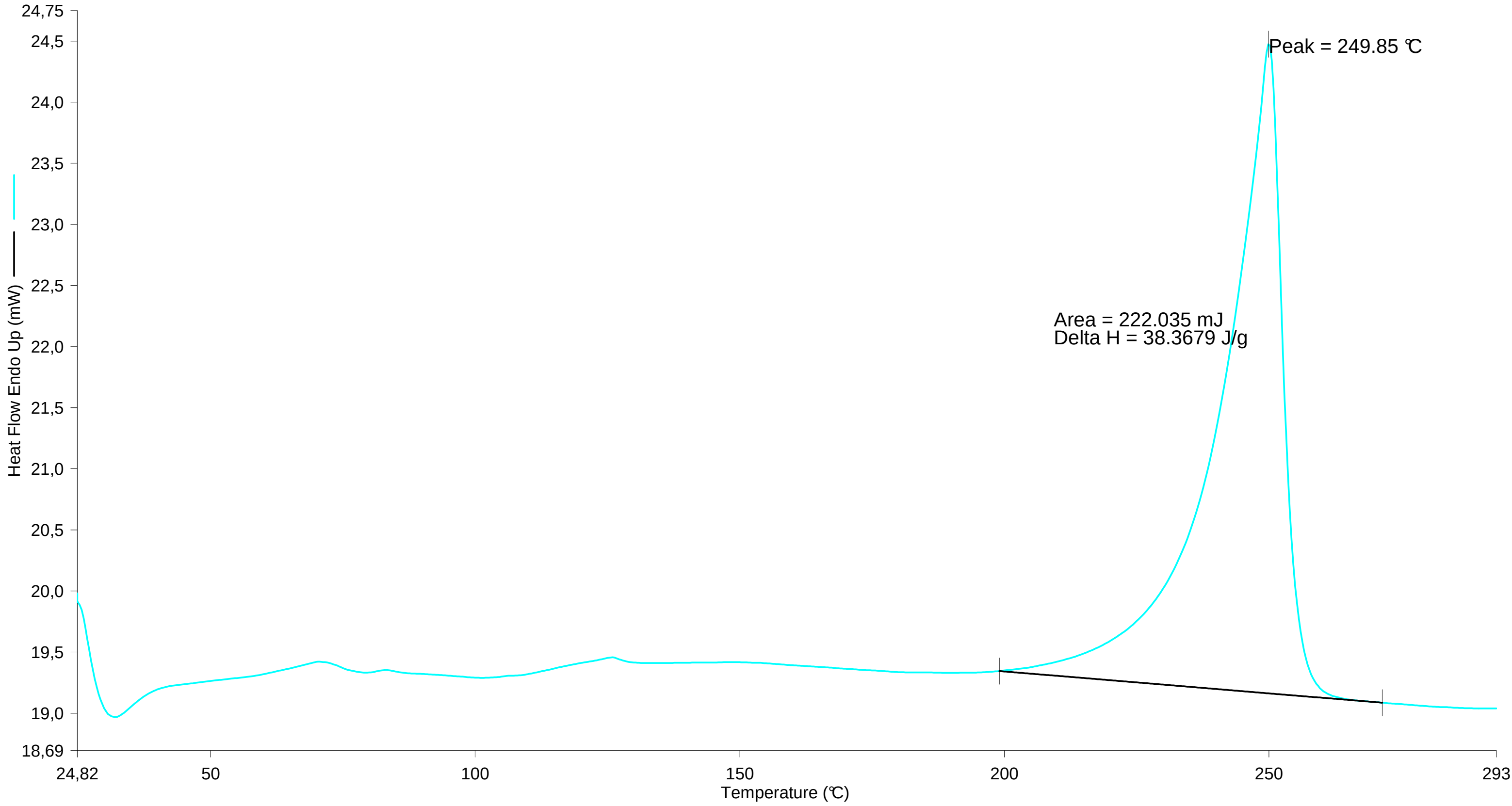
Comment:



10.03.2026 10:26:23

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2) Cool from 300.00℃ to 30.00℃ at 20.00℃/min
3) Heat from 30.00℃ to 300.00℃ at 20.00℃/min

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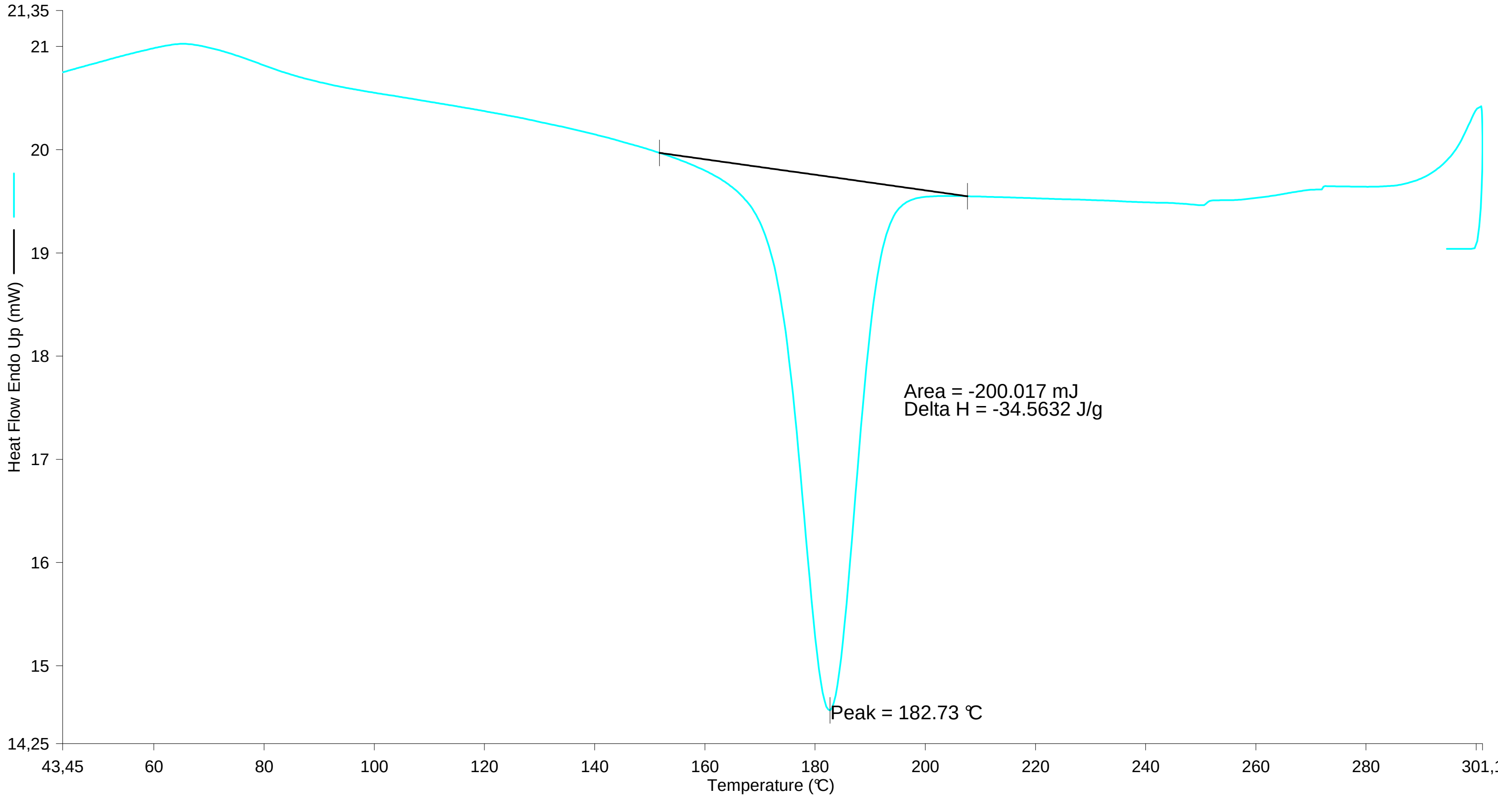
Sample ID: G8-PET
Sample Weight: 5.787 mg
Comment:



10.03.2026 10:27:39

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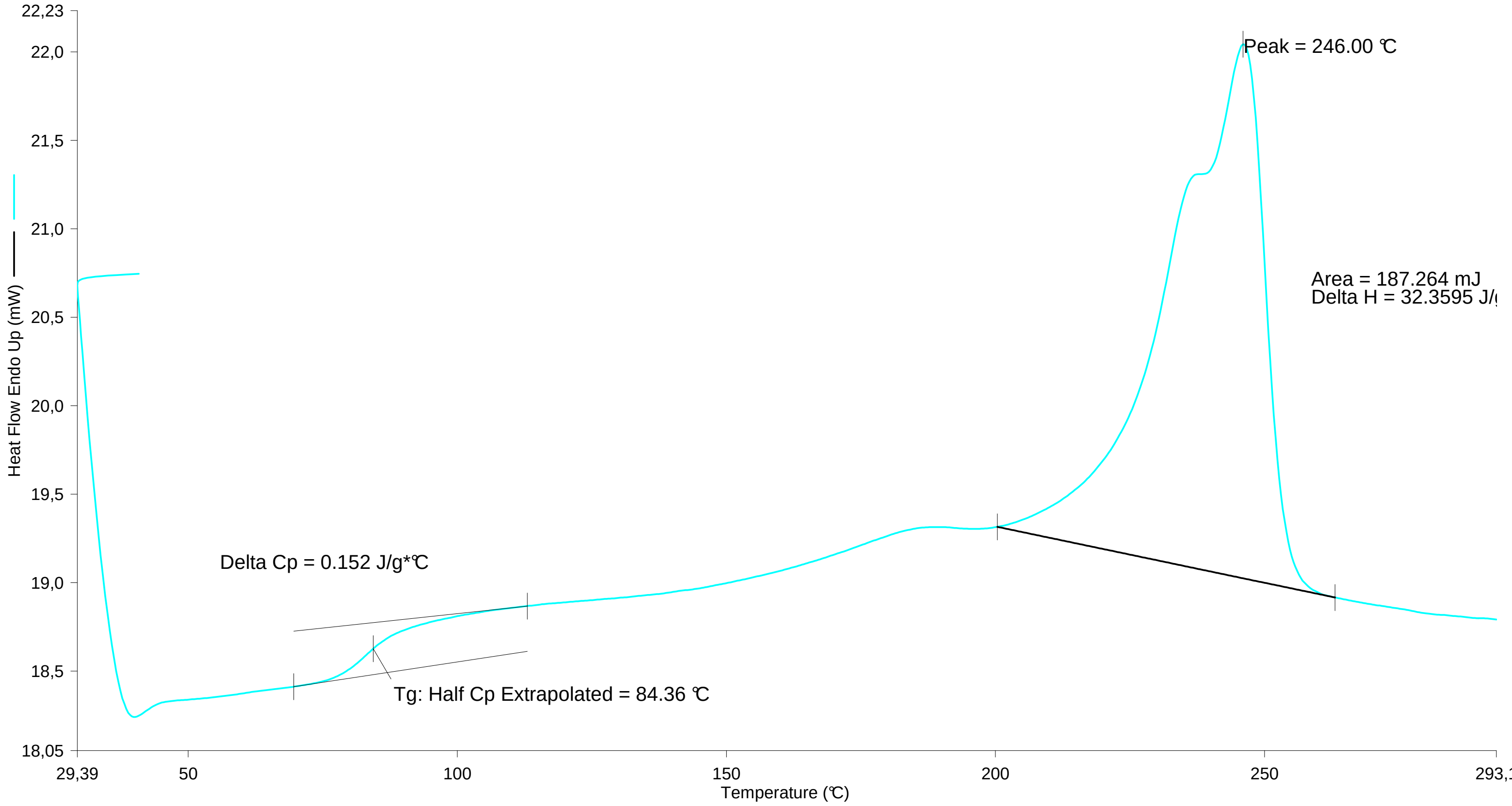
Sample ID: G8-PET
Sample Weight: 5.787 mg
Comment:



10.03.2026 10:29:27

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3) Heat from 30.00℃ to 300.00℃ at 20.00℃/min

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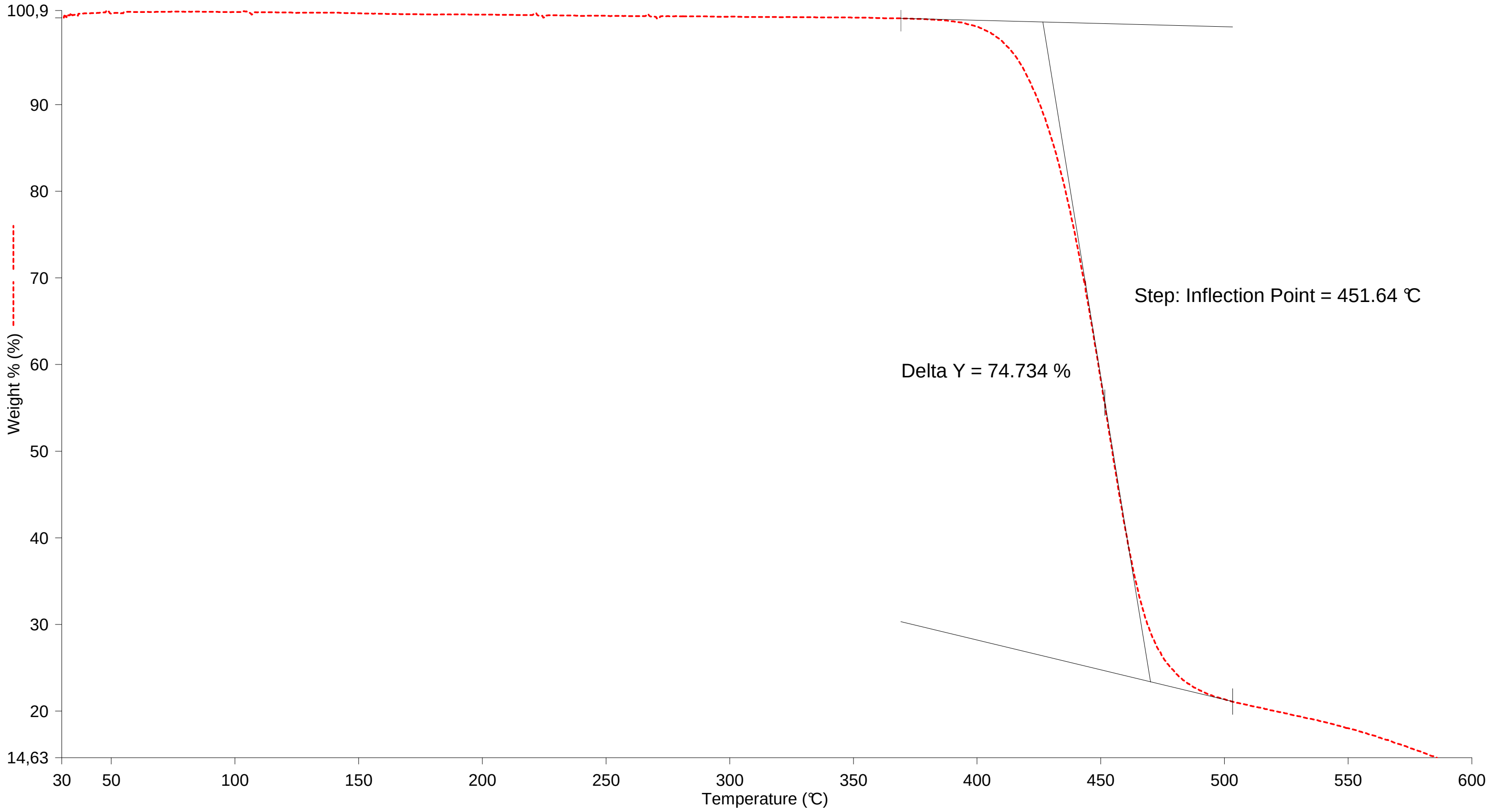
Sample ID: G8-PET
Sample Weight: 3.597 mg
Comment:



10.03.2026 10:42:27

1) Heat from 30.00℃ to 600.00℃ at 20.00℃/min

References

[1] Institut für Polymerchemie - Polymerpraktikum, Vorlesungsskript, Universität Stuttgart, **März 2025**.

[2] Y. Patil, J. Zhao, M. T. Puthiyaveettil, A. Louhichi, S. Rastogi, *Polymer* **2025**, *326*, 128333.

[3] In *Thermal Analysis of Polymeric Materials*, (Ed.: B. Wunderlich), Springer, Berlin, Heidelberg, **2005**, pp. 455–590.